Non-labelled Sequent Calculi of Public Announcement Expansions of **K45** and **S5**

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Outline

1. Sequent Calculus, Cut-elimination and **PAL**
2. Definition and Properties of G(**K45**) and G(**K45PAL**)
3. Definition and Properties of G(**S5**) and G(**S5PAL**)

Outline

Gerhard Gentzen

Untersuchungen über das logische Schließen*). I.

Von

Gerhard Gentzen in Göttingen.

Übersicht.

Die folgenden Untersuchungen beziehen sich auf den Bereich der Prädikatenlogik [bei H.-A.[1]) „engerer Funktionenkalkül" genannt]. Diese umfaßt solche Schlüsse, die in allen Teilen der Mathematik immerzu gebraucht werden. Was noch zu ihnen hinzukommt, sind Axiome und Schlußweisen, die man den einzelnen Zweigen der Mathematik selbst zurechnen kann, z. B.

G. Gentzen (1934). "Untersuchungen über das logische Schließen. I". *Mathematische Zeitschrift*. 39 (2): 176–210.

Sequent calculi for modal logics

$$\varphi_1, \varphi_2, \varphi_3, \cdots, \varphi_m \Rightarrow \psi_1, \psi_2, \psi_3, \cdots, \psi_n$$

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Public announcement logic (**PAL**)

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$$[\varphi]\psi$$

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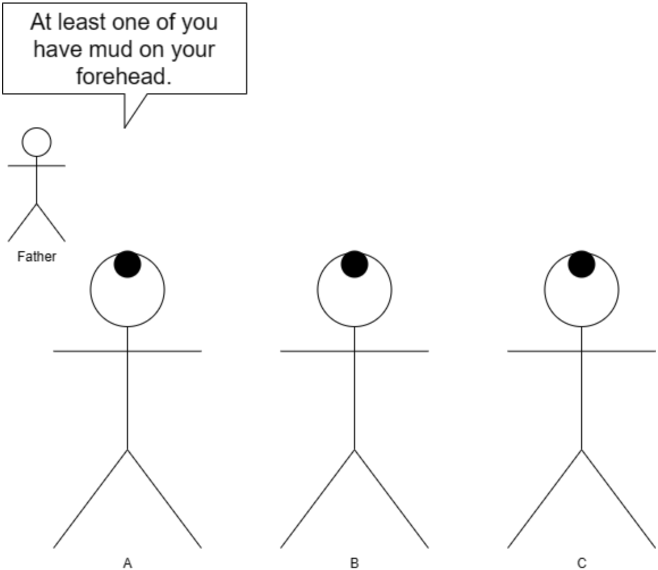
"If φ is true,
then after the public announcement of φ, ψ holds."

Model update: muddy children



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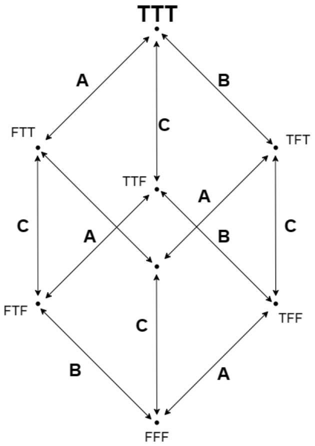
- *T*: muddy
- *F*: not muddy
- *p*: At least one of you have mud on your forehead.



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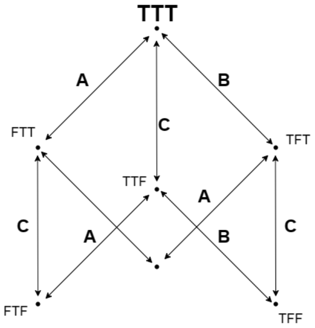

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  - ($x : \varphi$) Wu et al. (2022): No stack for announcements. Rule for them are from recursion axioms. Decidable.

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- Wising and Knapp (2022): non-labelled cut-free sequent calculus for action model logic based on **S4**.
- All papers above transform each of the recursion axioms into inference rules.

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Main Result of This Talk

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- While G(**S5PAL**) is not cut-free, it enjoys the (extended) subformula property as in (Takano 1992).

Outline

Syntax and Complexity

Syntax and Complexity

$$L \ni \varphi ::= p \mid \neg\varphi \mid \varphi \wedge \psi \mid \varphi \to \psi \mid K_a\varphi$$

$$L^+ \ni \varphi ::= p \mid \neg\varphi \mid \varphi \wedge \psi \mid \varphi \to \psi \mid K_a\varphi \mid [\varphi]\psi$$

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$$L^+ \ni \varphi ::= p \mid \neg\varphi \mid \varphi \wedge \psi \mid \varphi \to \psi \mid K_a\varphi \mid [\varphi]\psi$$

Complexity $c(\varphi)$ ($\varphi \in L^+$) (van Ditmarsch et al. 2007)

$$\begin{array}{rcl}
c(p) & := & 1, \\
c(\neg\varphi) & := & 1 + c(\varphi), \\
c(\varphi_1 \bullet \varphi_2) & := & 1 + \max(c(\varphi_1), c(\varphi_2)) \quad (\bullet \in \{\to, \wedge\}), \\
c(K_a\varphi) & := & 1 + c(\varphi), \\
c([\varphi]\psi) & := & (4 + c(\varphi)) \cdot c(\psi).
\end{array}$$

Closure $\mathtt{CL}(\varphi)$ for $\varphi \in L^+$ (van Ditmarsch et al. 2007)

$$
\begin{array}{rcl}
\mathtt{CL}(p) & := & \{p\} \\
\mathtt{CL}(\neg\varphi) & := & \mathtt{CL}(\varphi) \cup \{\neg\varphi\} \\
\mathtt{CL}(\varphi \bullet \psi) & := & \mathtt{CL}(\varphi) \cup \mathtt{CL}(\psi) \cup \{\varphi \bullet \psi\} \ (\bullet \in \{\rightarrow, \wedge\}) \\
\mathtt{CL}(K_a\varphi) & := & \mathtt{CL}(\varphi) \cup \{K_a\varphi\} \\
\mathtt{CL}([\varphi]p) & := & \mathtt{CL}(\varphi) \cup \{p, [\varphi]p\} \\
\mathtt{CL}([\varphi]\neg\psi) & := & \mathtt{CL}(\varphi) \cup \mathtt{CL}([\varphi]\psi) \cup \{[\varphi]\neg\psi\} \\
\mathtt{CL}([\varphi](\psi \bullet \gamma)) & := & \mathtt{CL}([\varphi]\psi) \cup \mathtt{CL}([\varphi]\gamma) \cup \{[\varphi](\psi \bullet \gamma)\} \ (\bullet \in \{\rightarrow, \wedge\}) \\
\mathtt{CL}([\varphi]K_a\psi) & := & \mathtt{CL}(K_a[\varphi]\psi) \cup \mathtt{CL}(\varphi) \cup \{[\varphi]K_a\psi\} \\
\mathtt{CL}([\varphi][\psi]\gamma) & := & \mathtt{CL}([\varphi \wedge [\varphi]\psi]\gamma) \cup \{[\varphi][\psi]\gamma\}
\end{array}
$$

For a set Ξ of formulas, we define $\mathtt{CL}(\Xi) := \bigcup_{\varphi \in \Xi} \mathtt{CL}(\varphi)$.

- $\mathtt{Sub}(\varphi) \subseteq \mathtt{CL}(\varphi)$,
  where $\mathtt{Sub}(\varphi)$ is the set of all subformulas of φ.

Frames, Models and Frame Classes

- A frame F is a tuple $(W, (R_a)_{a \in Ag})$ where W is a set of states and $R_a \subseteq W \times W$ is a binary relation on W.
- A model M is a tuple (F, V) where F is a frame and V is a function from Prop to $\mathcal{P}(W)$.
- $\mathbb{F}_{\mathbf{K45}}$: the class of all transitive and euclidean frames
- $\mathbb{F}_{\mathbf{S5}}$: the class of all reflexive, transitive and euclidean frames.

Semantics

Let $M = (W, (R_a)_{a \in Ag}, V)$ be a model and $w \in W$. The notion of φ being true at w in M (notation: $M, w \models \varphi$) is defined inductively as follows:

$$
\begin{array}{lll}
M, w \models p & \text{iff} & w \in V(p), \\
M, w \models \neg\varphi & \text{iff} & M, w \not\models \varphi, \\
M, w \models \varphi \wedge \psi & \text{iff} & M, w \models \varphi \text{ and } M \models \psi, \\
M, w \models \varphi \rightarrow \psi & \text{iff} & M, w \not\models \varphi \text{ or } M \models \psi, \\
M, w \models K_a\varphi & \text{iff} & \text{for all } v \in W, wR_a v \text{ implies } M, v \models \varphi, \\
M, w \models [\varphi]\psi & \text{iff} & M, w \models \varphi \text{ implies } M^\varphi, w \models \psi,
\end{array}
$$

where $M^\varphi = (W^\varphi, (R_a^\varphi)_{a \in Ag}, V^\varphi)$,
$W^\varphi = \{ w \in W \mid M, w \models \varphi \}$, $R_a^\varphi = R_a \cap (W^\varphi \times W^\varphi)$ and
$V^\varphi(p) = V(p) \cap W^\varphi$.

Hilbert Systems (van Ditmarsch et al. 2007)

All the Axioms and Rules of H(**K45**)	
(Taut)	all instances of propositional tautologies
(K)	$K_a(\varphi \to \psi) \to (K_a\varphi \to K_a\psi)$
(4)	$K_a\varphi \to K_aK_a\varphi$
(5)	$\neg K_a\varphi \to K_a\neg K_a\varphi$
(MP)	From $\varphi \to \psi$ and φ infer ψ
(Nec)	From φ infer $K_a\varphi$
Additional Axiom for H(S5)	
(T)	$K_a\varphi \to \varphi$
Additional Recursion Axioms for H(K45PAL) and H(S5PAL)	
([]at)	$[\varphi]p \leftrightarrow (\varphi \to p)$
([]$\wedge$)	$[\varphi](\psi \wedge \gamma) \leftrightarrow ([\varphi]\psi \wedge [\varphi]\gamma)$
([] $\to$)	$[\varphi](\psi \to \gamma) \leftrightarrow ([\varphi]\psi \to [\varphi]\gamma)$
([]$\neg$)	$[\varphi]\neg\psi \leftrightarrow (\varphi \to \neg[\varphi]\psi)$
([]K)	$[\varphi]K_a\psi \leftrightarrow (\varphi \to K_a[\varphi]\psi)$
([][])	$[\varphi][\psi]\gamma \leftrightarrow [\varphi \wedge [\varphi]\psi]\gamma$

Soundness and Completeness

Proposition (van Ditmarsch et al. 2007)

1. Hilbert systems H(**K45**) and H(**S5**) are sound and complete for $\mathbb{F}_{\mathbf{K45}}$ and $\mathbb{F}_{\mathbf{S5}}$, respectively.
2. Hilbert systems H(**K45PAL**) and H(**S5PAL**) are sound and complete for $\mathbb{F}_{\mathbf{K45}}$ and $\mathbb{F}_{\mathbf{S5}}$, respectively.

$$
\begin{array}{llll}
(c = & 25) & & [p][q]p \\
(c = & 10) & \leftrightarrow & [p \wedge [p]q]p \\
(c = & 7) & \leftrightarrow & ((p \wedge [p]q) \to p) \\
(c = & 4) & \leftrightarrow & ((p \wedge (p \to q))) \to p)
\end{array}
$$

Sequent Calculus $\mathbf{LK_0}$

- Axioms:

$$\frac{}{\varphi \Rightarrow \varphi}\ (\mathtt{id})$$

- Structural Rules:

$$\frac{\Gamma \Rightarrow \Delta}{\Gamma \Rightarrow \Delta, \varphi}\ (\Rightarrow w) \qquad \frac{\Gamma \Rightarrow \Delta}{\varphi, \Gamma \Rightarrow \Delta}\ (w \Rightarrow)$$

$$\frac{\Gamma \Rightarrow \Delta, \varphi, \varphi}{\Gamma \Rightarrow \Delta, \varphi}\ (\Rightarrow c) \qquad \frac{\varphi, \varphi, \Gamma \Rightarrow \Delta}{\varphi, \Gamma \Rightarrow \Delta}\ (c \Rightarrow)$$

- Logical Rules:

$$\frac{\Gamma \Rightarrow \Delta, \varphi_1 \quad \Gamma \Rightarrow \Delta, \varphi_2}{\Gamma \Rightarrow \Delta, \varphi_1 \wedge \varphi_2}\ (\Rightarrow \wedge) \qquad \frac{\varphi_i, \Gamma \Rightarrow \Delta}{\varphi_1 \wedge \varphi_2, \Gamma \Rightarrow \Delta}\ (\wedge \Rightarrow)$$

$$\frac{\varphi, \Gamma \Rightarrow \Delta}{\Gamma \Rightarrow \Delta, \neg\varphi}\ (\Rightarrow \neg) \qquad \frac{\Gamma \Rightarrow \Delta, \varphi}{\neg\varphi, \Gamma \Rightarrow \Delta}\ (\neg \Rightarrow)$$

$$\frac{\varphi, \Gamma \Rightarrow \Delta, \psi}{\Gamma \Rightarrow \Delta, \varphi \to \psi}\ (\Rightarrow\to) \qquad \frac{\Gamma \Rightarrow \Delta, \varphi \quad \psi, \Pi \Rightarrow \Sigma}{\varphi \to \psi, \Gamma, \Pi \Rightarrow \Delta, \Sigma}\ (\to\Rightarrow)$$

- Cut:

$$\frac{\Gamma \Rightarrow \Delta, \varphi \quad \varphi, \Pi \Rightarrow \Sigma}{\Gamma, \Pi \Rightarrow \Delta, \Sigma}\ (Cut)_\varphi$$

Sequent Calculus G(**K45**)

G(**K45**) expands $\mathbf{LK_0}$ with the following rule:

$$\frac{\Gamma, K_a\Gamma \Rightarrow K_a\Delta, \varphi}{K_a\Gamma \Rightarrow K_a\Delta, K_a\varphi}\ (K_{\mathbf{K45}}).$$

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$$\dfrac{\dfrac{\dfrac{K_a\varphi \Rightarrow K_a\varphi}{\varphi, K_a\varphi \Rightarrow K_a\varphi}\ (w \Rightarrow)}{K_a\varphi \Rightarrow K_aK_a\varphi}\ (K_{\mathbf{K45}})}{\Rightarrow K_a\varphi \to K_a\varphi K_a\varphi}\ (\Rightarrow\to) \qquad \dfrac{\dfrac{\dfrac{\dfrac{K_a\varphi \Rightarrow K_a\varphi}{\Rightarrow K_a\varphi, \neg K_a\varphi}\ (\Rightarrow \neg)}{\Rightarrow K_a\varphi, K_a\neg K_a\varphi}\ (K_{\mathbf{K45}})}{\neg K_a\varphi \Rightarrow K_a\neg K_a\varphi}\ (\neg \Rightarrow)}{\Rightarrow \neg K_a\varphi \to K_a\neg K_a\varphi}\ (\Rightarrow\to)$$

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- (*Cut*) can be eliminated (Shvarts 1989).

$$([\,]\,[\,]) \quad [\varphi][\psi]\gamma \leftrightarrow [\varphi \wedge [\varphi]\psi]\gamma$$

$$([]\,[])\quad [\varphi][\psi]\gamma \leftrightarrow [\varphi \wedge [\varphi]\psi]\gamma$$

$$\frac{\Gamma \Rightarrow \Delta, [\varphi \wedge [\varphi]\psi]\gamma}{\Gamma \Rightarrow \Delta, [\varphi][\psi]\gamma}\,(\Rightarrow []\,[]) \quad \frac{[\varphi \wedge [\varphi]\psi]\gamma, \Gamma \Rightarrow \Delta}{[\varphi][\psi]\gamma, \Gamma \Rightarrow \Delta}\,([]\,[] \Rightarrow)$$

Sequent Calculus G(**K45PAL**)

G(**K45PAL**) expands G(**K45**) with the following rules:

$$\frac{\varphi, \Gamma \Rightarrow \Delta, p}{\Gamma \Rightarrow \Delta, [\varphi]p} (\Rightarrow [\,]\,\mathtt{at}) \qquad \frac{\Gamma \Rightarrow \Delta, \varphi \quad p, \Pi \Rightarrow \Sigma}{[\varphi]p, \Gamma, \Pi \Rightarrow \Delta, \Sigma} ([\,]\,\mathtt{at} \Rightarrow)$$

$$\frac{\varphi, [\varphi]\psi, \Gamma \Rightarrow \Delta}{\Gamma \Rightarrow \Delta, [\varphi]\neg\psi} (\Rightarrow [\,]\,\neg) \qquad \frac{\Gamma \Rightarrow \varphi, \Delta \quad \Pi \Rightarrow \Sigma, [\varphi]\psi}{[\varphi]\neg\psi, \Gamma, \Pi \Rightarrow \Delta, \Sigma} ([\,]\,\neg \Rightarrow)$$

$$\frac{\Gamma \Rightarrow \Delta, [\varphi]\psi_1 \quad \Gamma \Rightarrow \Delta, [\varphi]\psi_2}{\Gamma \Rightarrow \Delta, [\varphi](\psi_1 \wedge \psi_2)} (\Rightarrow [\,]\wedge) \qquad \frac{[\varphi]\psi_i, \Gamma \Rightarrow \Delta}{[\varphi](\psi_1 \wedge \psi_2), \Gamma \Rightarrow \Delta} ([\,]\wedge \Rightarrow)$$

$$\frac{[\varphi]\psi, \Gamma \Rightarrow \Delta, [\varphi]\gamma}{\Gamma \Rightarrow \Delta, [\varphi](\psi \to \gamma)} (\Rightarrow [\,] \to) \qquad \frac{\Gamma \Rightarrow \Delta, [\varphi]\psi \quad [\varphi]\gamma, \Pi \Rightarrow \Sigma}{[\varphi](\psi \to \gamma), \Gamma, \Pi \Rightarrow \Delta, \Sigma} ([\,] \to\Rightarrow)$$

$$\frac{\varphi, \Gamma \Rightarrow \Delta, K_a[\varphi]\psi}{\Gamma \Rightarrow \Delta, [\varphi]K_a\psi} (\Rightarrow [\,]\,K) \qquad \frac{\Gamma \Rightarrow \Delta, \varphi \quad K_a[\varphi]\psi, \Pi \Rightarrow \Sigma}{[\varphi]K_a\psi, \Gamma, \Pi \Rightarrow \Delta, \Sigma} ([\,]\,K \Rightarrow)$$

$$\frac{\Gamma \Rightarrow \Delta, [\varphi \wedge [\varphi]\psi]\gamma}{\Gamma \Rightarrow \Delta, [\varphi][\psi]\gamma} (\Rightarrow [\,]\,[\,]) \qquad \frac{[\varphi \wedge [\varphi]\psi]\gamma, \Gamma \Rightarrow \Delta}{[\varphi][\psi]\gamma, \Gamma \Rightarrow \Delta} ([\,]\,[\,] \Rightarrow)$$

Complexity and Closure

Lemma

1. $c([\varphi]K_a\psi) > c(K_a[\varphi]\psi)$
2. $c([\varphi_1][\varphi_2]\psi) > c([\varphi_1 \wedge [\varphi_1]\varphi_2]\psi)$

Recall: Complexity $c(\varphi)$ ($\varphi \in L^+$)

$$\begin{array}{rcl} c(p) & := & 1, \\ c(\neg\varphi) & := & 1 + c(\varphi), \\ c(\varphi_1 \bullet \varphi_2) & := & 1 + \max(c(\varphi_1), c(\varphi_2)) \quad (\bullet \in \{\rightarrow, \wedge\}), \\ c(K_a\varphi) & := & 1 + c(\varphi), \\ c([\varphi]\psi) & := & (4 + c(\varphi)) \cdot c(\psi). \end{array}$$

Equipollence

Proposition

Let $\Lambda \in \{\mathbf{K45}, \mathbf{K45PAL}\}$.

1. If $\Gamma \Rightarrow \Delta$ is derivable in $G(\Lambda)$ then $\bigwedge \Gamma \rightarrow \bigvee \Delta$ is a theorem of $H(\Lambda)$.
2. If φ is a theorem of $H(\Lambda)$, then $\Rightarrow \varphi$ is derivable in $G(\Lambda)$.

Cut-elimination for G(**K45PAL**)

Cut Elimination of G(K45PAL)

If a sequent is derivable in G(**K45PAL**)
then it is derivable in G(**K45PAL**) w/o (*Cut*).

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(Cor.) Consistency for G(K45PAL)

The sequent $\Rightarrow$ is not derivable in G(**K45PAL**).

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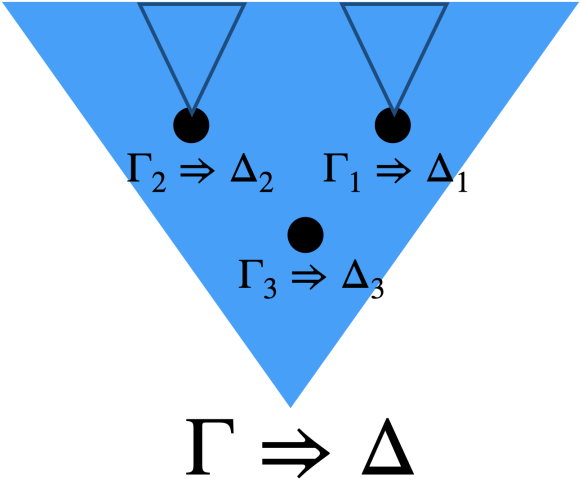

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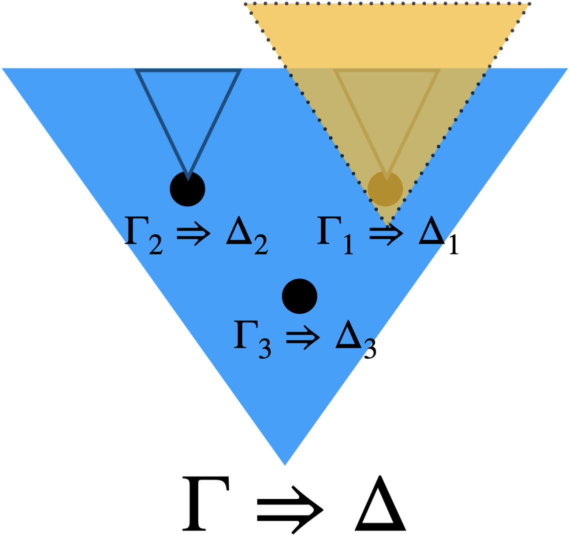

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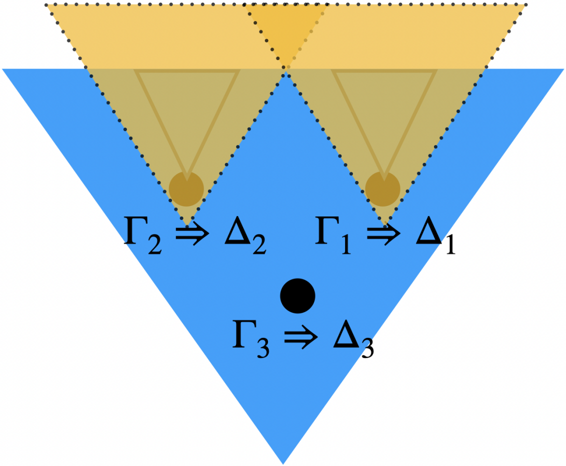

$$\Gamma \Rightarrow \Delta$$

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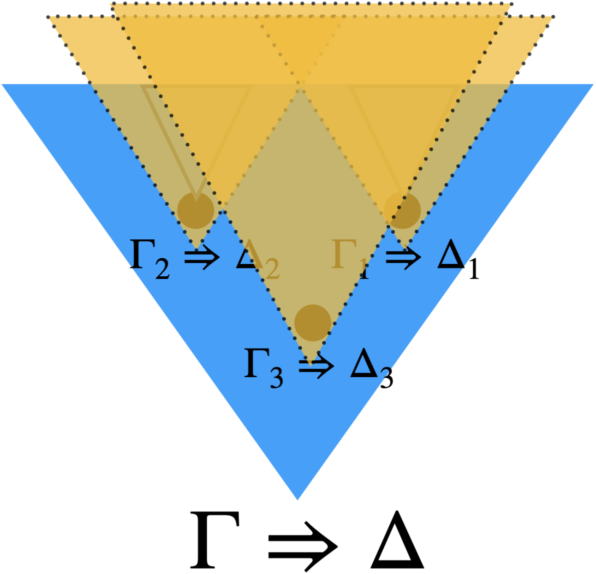

How Can We Eliminate Cut?

So, it suffices to eliminate the final applications of (Cut):

$$\mathcal{D} \equiv \dfrac{\begin{array}{c}\vdots\ \mathcal{L}\\ \Gamma \Rightarrow \Delta, \varphi\end{array} \quad \begin{array}{c}\vdots\ \mathcal{R}\\ \varphi, \Pi \Rightarrow \Sigma\end{array}}{\Gamma, \Pi \Rightarrow \Delta, \Sigma}\ (Cut).$$

Our proof is by induction on the lexicographic order of complexity $\mathtt{c}$ and weight $\mathtt{w}$ of $\mathcal{D}$:

- complexity $\mathtt{c}(\mathcal{D})$ is $\mathtt{c}(\varphi)$ of the cut-formula φ.
- weight $\mathtt{w}(\mathcal{D})$ is the number of all sequents in $\mathcal{L}$ and $\mathcal{R}$.

Typical case

- Suppose $\texttt{rule}(\mathcal{L})$ is $(\Rightarrow [\,]\,[\,])$ and $\texttt{rule}(\mathcal{R})$ is $([\,]\,[\,]\Rightarrow)$ where the cut-formula is principal in both rules:

$$\dfrac{\dfrac{\begin{array}{c}\vdots\ \mathcal{L}\\ \Gamma \Rightarrow \Delta, [\varphi \wedge [\varphi]\psi]\gamma\end{array}}{\Gamma \Rightarrow \Delta, [\varphi][\psi]\gamma}\ (\Rightarrow [\,]\,[\,]) \qquad \dfrac{\begin{array}{c}\vdots\ \mathcal{R}\\ [\varphi \wedge [\varphi]\psi]\gamma, \Pi \Rightarrow \Sigma\end{array}}{[\varphi][\psi]\gamma, \Pi \Rightarrow \Sigma}\ ([\,]\,[\,]\Rightarrow)}{\Gamma, \Pi \Rightarrow \Delta, \Sigma}\ (\mathit{Cut})_{[\varphi][\psi]\gamma}.$$

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- Suppose $\texttt{rule}(\mathcal{L})$ is $(\Rightarrow [\,][\,])$ and $\texttt{rule}(\mathcal{R})$ is $([\,][\,] \Rightarrow)$ where the cut-formula is principal in both rules:

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- We transform this derivation into:

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- By induction hypothesis, this (Cut) can be eliminated because $\mathrm{c}([\varphi][\psi]\gamma) > \mathrm{c}([\varphi \wedge [\varphi]\psi]\gamma)$.

Outline

Sequent Calculus G(**S5**)

G(**S5**) expands $\mathbf{LK_0}$ with the following rules:

$$\frac{K_a\Gamma \Rightarrow K_a\Delta, \varphi}{K_a\Gamma \Rightarrow K_a\Delta, K_a\varphi}\ (\Rightarrow K_{\mathbf{S5}}) \quad \frac{\varphi, \Gamma \Rightarrow \Delta}{K_a\varphi, \Gamma \Rightarrow \Delta}\ (K \Rightarrow).$$

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Proposition (Ohnishi and Matsumoto 1959)

$p \Rightarrow K_a\neg K_a\neg p$ is not derivable in G(**S5**) w/o (Cut).

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Subformula Property of G(**S5**) (Takano 1992)

If $\Gamma \Rightarrow \Delta$ is derivable in G(**S5**) then it is derivable in G(**S5**) with only analytic applications of (Cut).

Sequent Calculus G(**S5PAL**)

- G(**S5PAL**) expands G(**S5**) with the same rules in the case of G(**K45PAL**).

Proposition

Let $\Lambda \in \{\mathbf{S5}, \mathbf{S5PAL}\}$.

1. If $\Gamma \Rightarrow \Delta$ is derivable in G(Λ) then $\bigwedge \Gamma \to \bigvee \Delta$ is a theorem of H(Λ).
2. If φ is a theorem of H(Λ), then $\Rightarrow \varphi$ is derivable in G(Λ).

Extended Analytic Cut Property of G(**S5PAL**)

- A cut

$$\frac{\Gamma \Rightarrow \Delta, \varphi \quad \varphi, \Pi \Rightarrow \Sigma}{\Gamma, \Pi \Rightarrow \Delta, \Sigma}\ (\mathit{Cut})_\varphi$$

  is analytic if $\varphi \in \mathtt{CL}(\Gamma, \Pi, \Delta, \Sigma)$.
- Define G^{a}(**S5PAL**) as the same calculus as G(**S5PAL**) except that (*Cut*) is always analytic.

Non-Analytic Cut Elimination of G(**S5PAL**)

- If a sequent is derivable in G(**S5PAL**) then it is derivable in G^{a}(**S5PAL**).
- If $\Gamma \Rightarrow \Delta$ is derivable in G(**S5PAL**) then it is derivable in G(**S5PAL**) by a derivation which consists of $\mathtt{CL}(\Gamma, \Delta)$ alone.

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(Cor.) Consistency of G(**S5PAL**)

The sequent $\Rightarrow$ is not derivable in G(**S5PAL**).

Non-Analytic Cut Elimination of G(**S5PAL**)

- Goal: if a sequent is derivable in G(**S5PAL**) then it is derivable in G^{a}(**S5PAL**).
- We use the following lemma which is modified from Takano(1992)'s argument

Lemma

If the sequent

$$K_a\Gamma \Rightarrow K_a\Delta$$

is derivable in G^{a}(**S5PAL**) but with an application of $(\Rightarrow K_{\mathbf{S5}})$ for its lowest inference, then the sequent

$$K_a\Gamma \Rightarrow K_a(\Delta_\varphi), \varphi$$

is derivable in G^{a}(**S5PAL**) where $\Delta_\varphi := \Delta \setminus \{\varphi\}$.

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$\Longrightarrow$ The complexity c from van Ditmarsch et al. (2007):

$$\mathrm{c}([\varphi]\psi) := (4 + \mathrm{c}(\varphi)) \cdot \mathrm{c}(\psi).$$

plays an important syntactic role.

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- (Known) G(**K45PAL**) and G(**S5PAL**) are decidable.

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- (Known) G(**K45PAL**) and G(**S5PAL**) are decidable.
- If a sequent $\Gamma \Rightarrow \Delta$ in L is derivable in G(**K45PAL**) then it is also derivable in G(**K45**).

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- If a sequent $\Gamma \Rightarrow \Delta$ in L is derivable in G(**K45PAL**) then it is also derivable in G(**K45**).
- If a sequent in L is derivable in G(**S5PAL**) then it is also derivable in G(**S5**).

Further Direction

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- hypersequent calculus and bi-sequent calculi of **PAL**.
- sequent calculi for action model logic based on **S5**.
- sequent calculi for other extensions of non-classical epistemic logics with public announcements that also incorporate recursion axioms.